

ISR 70-03 Data for Calculations Based on DVS 1618



Bostik (Shanghai) Management Co., Ltd.

An Arkema company

www.bostik.com



Adhesive Mechanical Properties Required Data List

Adhesive Type <u>ISR 70-03</u>				
Mechanical Properties		Value	Unit	Data from
1	Tensile strength σ_{ch}	3.4	MPa	01_70-03 性能检测报告_BETC-HJ-2015-M-00006
2	Static Shear strength τ_B (Char. value)	1.91	MPa	02_ISR70-03 剪切强度测试
3	Fatigue shear strength τ_B , fatigue	0.30	MPa	03_70-03 疲劳试验检测报告_SAT14CB00734
4	Tear strength	18	N/mm	01_70-03 性能检测报告_BETC-HJ-2015-M-00006
5	Static shear modulus G_s	1.2	MPa	E modulus(Es) calculated
6	Static E modulus E_s	3.6	MPa	04_70-03 拉伸模量检测报告_Z1706016800
7	Dynamic shear modulus G_D	4.3	MPa	05_70-03 冲击试验
8	Poisson ratio ν	0.45	/	同种材质, 借鉴 ISR 70-08 AP DIN6701 M9
9	Elongation at break ϵ	260	%	01_70-03 性能检测报告_BETC-HJ-2015-M-00006
10	Allowable strain, Tension/compression extraordinary	30	%	经验值
11	Allowable strain, shear extraordinary	60	%	经验值
12	Allowable strain, Tension/compression fatigue	20	%	经验值
13	Allowable strain, shear fatigue	40	%	经验值
14	Coefficient of thermal expansion	180E-6	1/°C	06_70-03 线性热膨胀系数检测报告 _A2170021539101C
15	Density	1.4	g/cm ³	01_70-03 性能检测报告_BETC-HJ-2015-M-00006
Reduction factor acc. DVS 1618		Value	Unit	
16	Reduction factor for the strength of the adhesive due to difference of test temperature and the ambient temperature(50°C), f_T	0.84	/	07_70-03 不同温度下的剪切强度测试
17	Reduction factor for medium flow, f_M	0.88	/	08_70-08 AP 耐酸碱测试(剪切)
18	Reduction factor for different geometry of specimens, f_G	0.71	/	11_粘接厚度对 ISR7003 剪切强度的影响
19	Reduction factor for static long term influence, f_L (including creep and ageing)	0.94	/	09_ISR7003/ISP7003sskf/ISR7008AP 湿热老化测试 10_SMP 抗蠕变测试
20	Reduction factor for dynamic load (impact or fatigue), f_D	0.87	/	03_70-03 疲劳试验检测报告_SAT14CB00734

Note: All A1/A2 class adhesives according to DIN 6701 should provide this data list (need include the original test report as appendix) to BST for strength calculation according to DVS 1618.

No.16	Reduction factor for the strength of the adhesive due to difference of test temperature and the ambient temperature(50°C), f_T				
Temp/°C	-40	0	23	50	90
LSS/MPa	2.6	2.2	1.9	1.6	1.3
condition			f_T		
50 °C			0.84 (1.6/1.9)		

No.17	Reduction factor for medium flow, f_M	
Condition	Before aging	After aging
LSS/MPa	2.5	2.2
condition		f_M
Medium resistance		0.88 (2.2/2.5)

No.18	Reduction factor for different geometry of specimens, f_G			
Thickness/mm	5.0	8.0	15.0	20.0
LSS/MPa	2.1	1.8	1.5	1.4
condition		f_G		
15.0mm		0.71 (1.5/2.1)		

No.19	Reduction factor for static long term influence, f_L (including creep and ageing)			
condition	Before heat-humidity	After heat-humidity	Before creep	After creep
LSS/MPa	1.7	1.6	2.4	2.4
f_{LA}		f_{LC}		
0.94 (1.6/1.7)		1.00 (2.4/2.4)		

No.20	Reduction factor for dynamic load (impact or fatigue), f_D	
Condition	Before cycle	After cycle
LSS/MPa	2.64	2.30
condition		f_D
Fatigue resistance		0.87 (2.30/2.64)